

Large-Scale BGP Routing Anomaly Detection Based on Graph Attention Auto-Encoder

Zekang Wang¹, Fuxiang Yuan¹, Han Qiu¹, Yan Liu¹, and Xiangyang Luo¹

Abstract—The Border Gateway Protocol (BGP) enables the exchange of routing information between Autonomous Systems (ASes). The lack of routing verification makes it susceptible to routing anomalies with major impact. Detecting large-scale routing anomalies is crucial for monitoring network operation status. However, existing methods are affected by interference from non-important ASes and insufficient labeled data, necessitating improvements in detection performance. In this paper, a large-scale BGP routing anomaly detection method based on a graph attention auto-encoder is proposed. First, normal BGP data is collected to construct AS-level topology graphs. Then, the graph attention auto-encoder model is pre-trained on topology graphs, where an attention mechanism is introduced to assign higher weights to important ASes. Next, the AS-level topology graph sequence is extracted from BGP data, and the pre-trained model is utilized to obtain AS embeddings. Finally, the AS embedding is classified to detect large-scale BGP routing anomalies based on multi-layer perceptron. The large-scale BGP routing anomaly event dataset is constructed using 4,577,384 routing announcements, and extensive experiments are conducted. The results show that the proposed method has significant advantages. Compared with typical methods *BSVM*, *BLSTM*, *BMLP*, *BGNN*, *BGIN*, and *BGAE*, the detection accuracy increased by 14.47%, 3.73%, 8.11%, 2.64%, 5.99%, and 1.04%, respectively.

Index Terms—BGP routing anomaly detection, graph attention auto-encoder, autonomous system, graph representation learning.

I. INTRODUCTION

THE INTERNET consists of over 80,000 Autonomous Systems (ASes), and ASes exchange routing information via the Border Gateway Protocol (BGP). However, BGP was not designed with a routing trusted verification

mechanism [1], [2], which makes the Internet susceptible to routing anomalies such as network outages [3] caused by route leaks [4], prefix hijacking [5], and configuration errors [6]. According to the impact scope of anomalies, they can be categorized into small-scale and large-scale routing anomalies [7], in which large-scale routing anomalies seriously impact network reachability [8]. Large-scale routing anomaly detection is of great significance in monitoring Internet operation status and maintaining network stability [9].

Machine learning has been widely applied in large-scale BGP routing anomaly detection for years, achieving many outstanding scientific achievements. Traditional methods based on machine learning extract anomaly detection features manually from routing information tables and UPDATE messages, including statistical features [10], [11] (e.g., the number of UPDATE messages, the average length of AS_PATH) and graph features [12], [13] (e.g., clustering coefficient, centrality). The machine learning model is trained to detect large-scale BGP routing anomalies. However, the process of manually extracting and selecting features relies on domain knowledge. With the development of deep learning, the emergence of automatic feature extraction methods is expected to solve such problems. In these methods, graph representation learning [14] can effectively encode high-dimensional sparse graph structure data into low-dimensional dense vectors [15], while preserving structural information in graphs, providing a new way for BGP routing anomaly detection. In recent years, some studies have attempted to apply graph representation learning to large-scale BGP routing anomaly detection tasks. Based on graph convolutional neural network [16], graph isomorphism network [17], and graph auto-encoder [18], the information of neighbor nodes is aggregated to update AS embeddings, and large-scale BGP routing anomaly is detected by classifying the AS embedding.

Existing large-scale BGP routing anomaly detection methods based on graph representation learning have made initial progress but still have some problems. Significant differences exist in the possibility of large-scale BGP routing anomalies caused by different ASes [19]. However, existing typical detection methods aggregate features of neighbor ASes indiscriminately, either by averaging or summing, to update AS embeddings, resulting in anomaly detection facing interference from many non-important or even irrelevant ASes. In addition, existing methods typically train detection models in a supervised manner, which requires a large amount of labeled BGP data. However, BGP anomaly data is insufficient. These

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problems affect the effectiveness of large-scale BGP routing anomaly detection.

To solve the above problems, we propose a novel method based on a graph attention auto-encoder for detecting large-scale BGP routing anomalies. The attention mechanism is used to assign higher weights to important ASes when aggregating neighbor node information, minimizing the interference of non-important ASes on anomaly detection. The reliance on labeled data is reduced by pre-training the graph attention auto-encoder model. The main contributions of this paper are as follows.

- *A framework for detecting large-scale BGP routing anomalies using a method based on a graph attention auto-encoder is proposed.* The proposed method learns graph representations of Internet AS-level topologies based on a graph attention auto-encoder and utilizes a multi-layer perceptron to classify the AS embedding, achieving anomaly detection. Our method also provides a list of possible anomaly origin ASes, narrowing down the scope of anomaly origin AS localization.
- *The attention mechanism is introduced into the graph auto-encoder for large-scale BGP routing anomaly detection.* When aggregating node features of neighbor ASes, the attention mechanism evaluates the importance of ASes and assigns weights to them, paying more attention to important ASes that may cause large-scale BGP routing anomalies. This mechanism reduces interference from non-important ASes and improves detection accuracy.
- *Normal behavior of BGP routing is learned based on unsupervised learning.* The graph attention auto-encoder model is pre-trained on normal BGP data to learn routing behavior patterns, reducing the dependence on labeled routing data. The model effectively captures the proximity and hierarchy properties of Internet AS-level topologies, which represent the AS routing roles.

The remainder of this paper is organized as follows. Section II discusses the related work of large-scale BGP routing anomaly detection. Section III elaborates on the proposed method based on a graph attention auto-encoder, detailing the framework, model architecture, and effectiveness analysis of the method. Section IV validates the effectiveness of the proposed method and compares it with existing typical detection methods. Finally, the conclusion summarizes our findings and outlines future research directions.

II. RELATED WORK

We introduce the definition and classification of large-scale BGP routing anomalies, and analyze existing typical methods for detecting large-scale BGP routing anomalies.

A. Large-Scale BGP Routing Anomaly

BGP is the most widely used inter-domain routing protocol [20], but it has obvious security flaws in design [21], [22]. This protocol lacks an effective inherent mechanism to verify the authenticity and validity of route announcements. As a result, the Internet becomes vulnerable to routing anomalies, such as network outages caused by configuration errors or

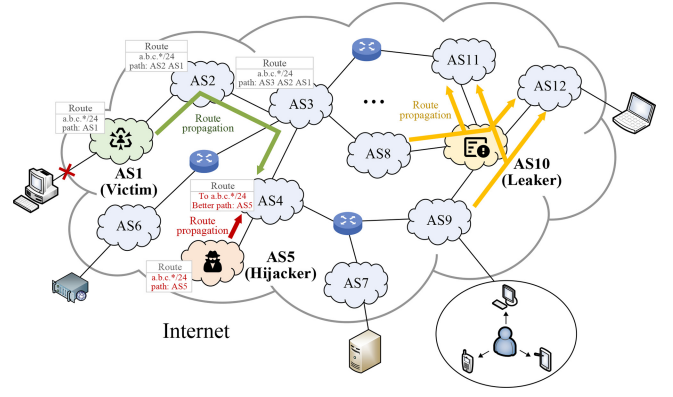


Fig. 1. **Illustration of large-scale BGP routing anomalies.** (1) Route leaks: Routes from AS8 and AS9 are propagated to unintended ASes, including AS11 and AS12. (2) Prefix hijacking: AS1 owns the network prefix a.b.c.*24, but the attacker AS5 illegitimately announces a route containing this network prefix. According to the BGP shortest path principle, the traffic is redirected to AS5.

malicious attacks [23]. According to the scope of impact, these anomalies can be divided into small-scale and large-scale routing anomalies. Large-scale routing anomalies [8] usually cause a large number of IP address prefixes to be unreachable, and the anomalies can be observed by most routing collectors, posing a serious threat to inter-domain routing security.

Common large-scale BGP routing anomalies include route leaks [24] and prefix hijacking [25], as shown in Fig. 1. Route leaks are typically caused by misconfigurations made by network operators [26], where legitimate routing paths are exposed to unintended ASes [27], putting critical Internet infrastructure at risk. Prefix hijacking refers to an attack where the attacker forges network reachability information and AS path information to announce that they own the victim's IP address prefix [28], deceiving BGP routers to select incorrect paths and redirecting network traffic to other ASes [29], thereby stealing user sensitive data [30]. Detecting large-scale BGP routing anomalies is of great importance for protecting critical Internet infrastructure and ensuring the security of user data [31].

B. Large-Scale BGP Routing Anomaly Detection Methods

Anomaly detection is one of the important branches of data mining. Its goal is to identify abnormal data, which is widely used in different scenarios [32], [33], [34], [35], [36], [37], [38], [39]. Early large-scale BGP routing anomaly detection methods include time series analysis [40], statistical pattern recognition [41], and reachability test [42]. In recent years, machine learning has shown good performance in large-scale BGP routing anomaly detection [43]. Machine learning-based detection methods are currently the mainstream approach. These methods focus on how to extract features effectively. Currently, there are two approaches for feature extraction: manual extraction and automatic extraction based on graph representation learning.

1) *Methods Based on Manual Feature Extraction*: Traditional methods detect large-scale BGP routing anomalies by manually extracting features. The extracted features can

be mainly categorized into two types: statistical features and graph features.

When large-scale BGP routing anomalies occur, relevant features of UPDATE messages change significantly. Al-Rousan and Trajković [10] extracted 17 statistical features, including the number of UPDATE messages, the announcement interval time, and the average length of AS_PATH. These features were selected using fisher score and mRMR algorithms, and a support vector machine classifier was trained to detect large-scale BGP routing anomalies. Dong et al. [11] extracted 33 statistical features regarding AS prefixes, peers, and paths. A long short-term memory model combined with a self-attention mechanism was utilized to detect large-scale BGP routing anomalies, in order to reduce the impact of noise in BGP data.

Graph features reflect overall structural changes in Internet AS-level topology when large-scale BGP routing anomalies occur. Sanchez et al. [12] compared large-scale BGP routing anomaly detection performance of graph features such as centrality, clustering coefficient, average degree of neighbor nodes, and maximum connected subgraph size using a multi-layer perceptron. Huang et al. [13] extracted degree centrality and closeness centrality features of ASes, achieving real-time detection of large-scale BGP routing anomalies by analyzing changes in AS centrality.

However, manual feature extraction and selection can easily lead to feature redundancy or the omission of important features. In addition, feature design is time-consuming and costly, which affects the accuracy of large-scale BGP routing anomaly detection.

2) *Methods Based on Graph Representation Learning for Feature Extraction:* Graph representation learning [14] can represent a graph as a low-dimensional vector while retaining most network structure information. The features can be automatically extracted by inputting the graph and a small number of initial node features into a graph representation learning model. In recent years, some studies have made progress in large-scale BGP routing anomaly detection based on graph representation learning.

Hoarau et al. [16] first designed a graph neural network model suitable for large-scale BGP routing anomaly detection. By aggregating the embeddings of neighbor ASes averagely, AS embeddings are iteratively updated without manual feature extraction. Then, AS embeddings are classified based on a multi-layer perceptron to achieve large-scale BGP routing anomaly detection. Latif et al. [17] utilized the number of UPDATE messages sent and received by ASes as the initial node features and applied a graph isomorphism network model to automatically extract deep features of AS-level topologies for large-scale BGP routing anomaly detection. This model employs a specialized information aggregation mechanism, which updates AS embeddings by summing the embeddings of neighbor ASes and the ASes themselves, allowing for a more accurate distinction between similar AS-level topologies. Hoarau et al. [18] utilized a graph auto-encoder to learn graph representations of AS-level topologies. The graph convolutional network is used as the encoder. The AS embeddings are updated by aggregating the information from neighbor ASes

averagely. During the training process, the error for all ASes is computed to update model parameters, capturing routing behavior patterns of different ASes.

In summary, while large-scale BGP routing anomaly detection methods based on graph representation learning have made initial progress, they still have some problems. Existing typical detection methods assign equal weights to all neighbor ASes when aggregating their information to update AS embeddings, assuming that different ASes causing large-scale BGP routing anomalies have the same possibility. However, ASes that trigger large-scale BGP routing anomalies are usually important nodes with larger network scales and significant network positions [19]. The existing method aggregates features of neighbor ASes by averaging or summing, which ignores the differences in importance between different ASes. As a result, anomaly detection faces interference from many non-important, even irrelevant ASes. Furthermore, BGP anomaly data samples are insufficient, and it is not guaranteed that all data can be accurately labeled. This makes it challenging to obtain a large amount of accurately labeled BGP anomaly data required for supervised learning. These problems affect the effectiveness of anomaly detection.

III. PROPOSED METHOD

To solve the problems of interference from non-important ASes and the insufficiency of labeled BGP anomaly data, we propose a large-scale BGP routing anomaly detection method based on a graph attention auto-encoder.

A. Overview

Considering that large-scale BGP routing anomalies are primarily characterized by significant changes in the existence of AS links [8], we represent the Internet AS-level topology as an undirected graph $G = (V, E, X)$. Here, $V = \{v_1, v_2, \dots, v_N\}$, where v_i denotes an AS and N is the number of ASes. $E = \{e_1, e_2, \dots, e_M\}$, e_j indicates an edge between ASes, and M denotes the number of edges. $X = \{x_1, x_2, \dots, x_N\}$ represents the initial node features of ASes, and x_k indicates the number of IP prefixes announced by an AS, reflecting its network size and connectivity capability. The notations used in the paper are summarized in Tab. I.

The overview architecture of our proposed framework is shown in Fig. 2, which consists of two stages. In the graph attention auto-encoder model pre-training stage, the model is pre-trained in an unsupervised manner on AS-level topologies without anomalies to learn normal BGP routing behaviors. In the large-scale BGP routing anomaly detection based on multi-layer perceptron stage, the encoder from the pre-trained graph attention auto-encoder model is utilized to generate AS embeddings of the AS-level topology graph sequence. The AS embedding is classified using a multi-layer perceptron to detect large-scale BGP routing anomalies. For a specific AS, if an anomaly is detected, the result $\hat{y} = 1$ is output. Otherwise, $\hat{y} = 0$.

B. Framework

The main steps of the proposed framework are as follows.

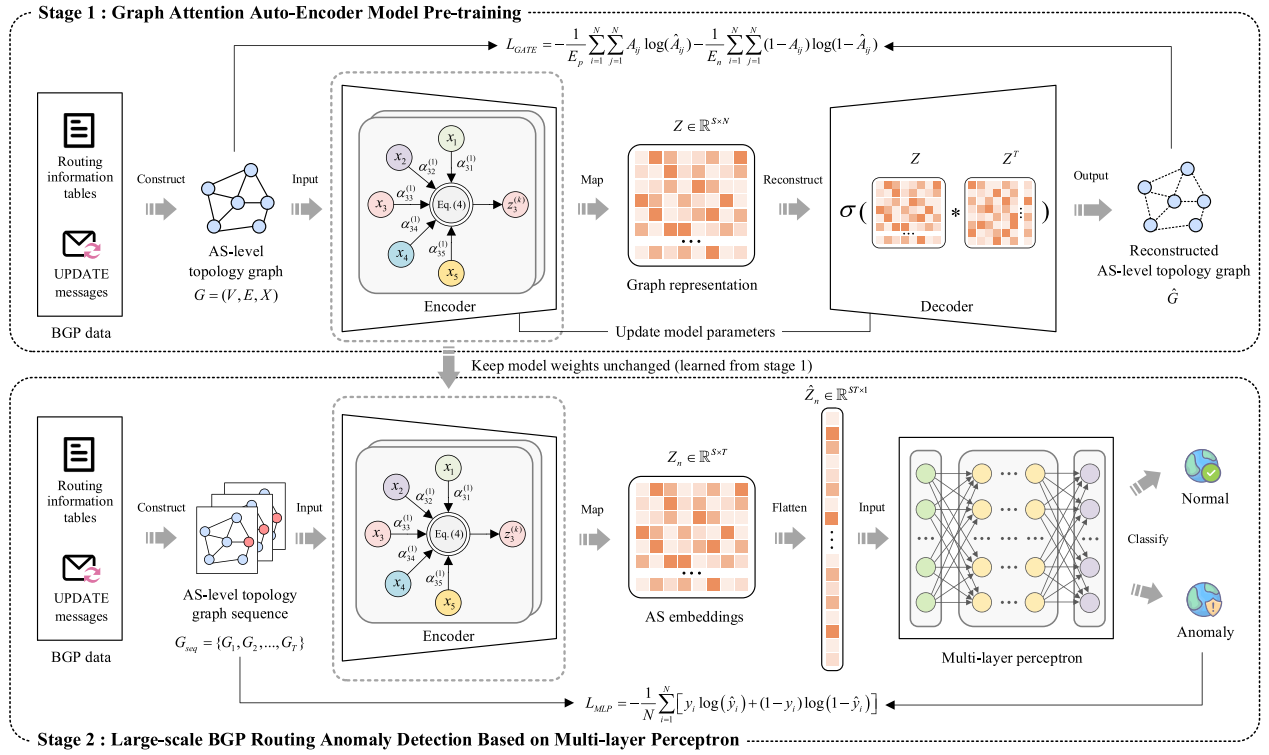


Fig. 2. Overview of our proposed detection framework. The framework consists of two stages: graph attention auto-encoder model pre-training and large-scale BGP routing anomaly detection based on multi-layer perceptron.

Stage 1: Graph attention auto-encoder model pre-training.

Step 1: Construct Internet AS-level topology graphs. Routing information tables and UPDATE messages, which do not involve large-scale BGP routing anomalies, are collected, and AS paths are extracted. Considering path information may contain erroneous data, we clean it. Reserved ASes have special purposes and sometimes appear in AS paths, which are eliminated. There are many cases where the same ASes are added to AS paths. Duplicate ASes are removed. Internet Exchange Points (IXPs) are excluded because they do not participate in the routing decision process. Operation and maintenance personnel may have routing configuration errors, resulting in self-loops in AS paths, so we filter them out. On this basis, AS-level topology graphs in normal state are constructed.

Step 2: Obtain graph representations of AS-level topologies. AS-level topology graphs are input into the encoder of the constructed graph attention auto-encoder model. Stacked graph attention layers encode AS-level topology graphs, mapping high-dimensional graph structure data to a low-dimensional representation space. The attention mechanism is introduced to dynamically adjust the weights of AS embeddings within AS neighbors, focusing on important ASes that may be involved in large-scale BGP routing anomalies. The AS embeddings are iteratively updated by aggregating node embeddings of neighbor ASes, outputting graph representations of AS-level topologies $Z \in \mathbb{R}^{S \times N}$.

Step 3: Reconstruct adjacency matrices of AS-level topology graphs. The decoder of the constructed graph

TABLE I
LIST OF NOTATIONS

Notation	Description
G	Internet AS-level topology graph, $G = (V, E, X)$.
G_{seq}	AS-level topology graph sequence, $G_{seq} = \{G_1, G_2, \dots, G_T\}$.
T	The number of graphs in sequence, $T = G_{seq} $.
V	The set of ASes, $V = \{v_1, v_2, \dots, v_N\}$.
N	The number of ASes, $N = V $.
E	The set of edges between ASes, $E = \{e_1, e_2, \dots, e_M\}$.
M	The number of edges, $M = E $.
X	Initial node features of ASes, $X = \{x_1, x_2, \dots, x_N\}$.
A	Adjacency matrix of Internet AS-level topology graph, $A_{ij} = 1$ if edge (v_i, v_j) exists, else 0.
$\hat{A}$	Reconstructed adjacency matrix of Internet AS-level topology graph output by the decoder.
z_i	The node embedding of AS i .
S	The dimension of AS embedding.
Z	The graph representation of Internet AS-level topology, $Z = \{z_1, z_2, \dots, z_N\} \in \mathbb{R}^{S \times N}$.
Z_n	The node embeddings for AS n of the AS-level topology graph sequence, $Z_n = \{z_n^1, z_n^2, \dots, z_n^T\} \in \mathbb{R}^{S \times T}$.
L	The depth of the encoder.
$\alpha_{ij}^{(k)}$	Attention coefficient between AS i and AS j in k -th encoder layer.
H	The number of attention heads.
t_0	The moment when an anomaly event occurs.
$\hat{y}$	The detection result of large-scale BGP routing anomaly.

attention auto-encoder model receives graph representations generated by the encoder, and maps them back to the original data space, reconstructing the adjacency matrix of the AS-level

topology graph. The model is trained by minimizing the error between the real and reconstructed adjacency matrices, updating the model weights to effectively learn graph representations of AS-level topologies under normal BGP routing behavior.

Stage 2: Large-scale BGP routing anomaly detection based on multi-layer perceptron.

Step 1: Construct a graph sequence of Internet AS-level topologies. Routing information tables and UPDATE messages within the detection period are collected from multiple vantage points deployed across the Internet. The AS paths are extracted and preprocessed to remove erroneous data, including reserved ASes, duplicate ASes, IXPs, and self-loops. At different timestamps, AS-level topology graphs are extracted from the preprocessed path information, forming a sequence of graphs $G_{seq} = \{G_1, G_2, \dots, G_T\}$, where the number of topology graphs is T .

Step 2: Obtain node embeddings of ASes in graph sequence of AS-level topologies. The graph sequence is input into the encoder of the pre-trained graph attention auto-encoder model. During pre-training, this model has learned normal BGP routing behavior patterns. Therefore, the weights of the encoder are kept unchanged. The attention mechanism is utilized to assign higher weights to important ASes, generating AS embeddings $Z_n = \{z_n^1, z_n^2, \dots, z_n^T\} \in \mathbb{R}^{S \times T}$ of the AS-level topology graph sequence G_{seq} , where z_n^i is the node embedding of AS n in the AS-level topology graph G_i .

Step 3: Detect large-scale BGP routing anomalies using a multi-layer perceptron. The AS embedding Z_n is flattened into one-dimensional vector $\hat{Z}_n$. When large-scale BGP routing anomalies occur, ASes exhibit abnormal routing behavior, and the AS embeddings generated by the pre-trained graph attention auto-encoder model will significantly differ from the embeddings in normal state. Therefore, a multi-layer perceptron is employed to classify these embeddings. The AS embeddings are passed through the input, hidden, and output layers, learning complex mapping relationships between features. This enables the detection of large-scale BGP routing anomalies, with the output of final detection results $\hat{y}$.

C. Graph Attention Auto-Encoder Model Pre-training

The proposed method learns from a large amount of unlabeled BGP data without anomalies, capturing normal routing behavior patterns. The graph attention auto-encoder model consists of an encoder and a decoder. The architecture of the encoder and decoder, as well as the loss function, are described as follows.

1) *Encoder Architecture*: The encoder comprises two graph attention layers, as shown in Fig. 3. The first graph attention layer transforms initial AS features into AS embeddings that contain network structure information. The second graph attention layer updates AS embeddings and learns more complex, deeper features. ReLU is used as the activation function between two neural network layers.

The process of aggregating AS neighbors' features based on the graph attention layer is shown in Fig. 4. First, since the dimension of initial node features is small, a learnable weight

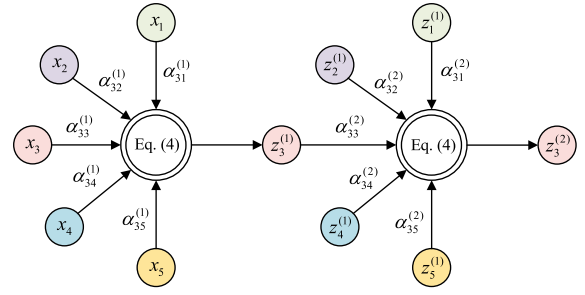


Fig. 3. The architecture of the encoder. The encoder computes coefficients between ASes using an attention mechanism, and iteratively updates AS embeddings.

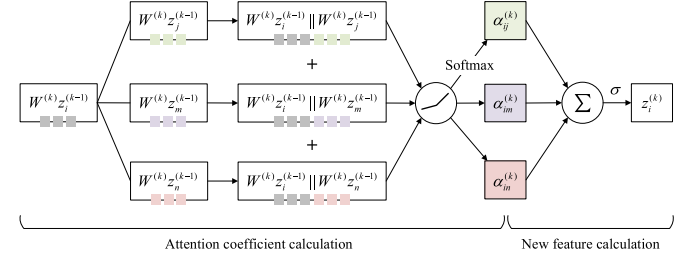


Fig. 4. The process of aggregating AS neighbors' features based on the graph attention layer. It consists of two parts: attention coefficient calculation and new feature calculation.

matrix is used for feature enhancement. Then, the correlation coefficient is obtained by concatenating and mapping the features of AS itself and neighbor nodes. Next, the correlation coefficient is normalized to compute the attention coefficient. Finally, the features of neighboring ASes are weighted and summed according to the attention coefficients. The new AS embedding is obtained after being processed by a nonlinear activation function.

The features of AS i and its neighbor AS j are concatenated and linearly transformed using a learnable weight matrix $\mathbf{W}^{(k)}$. After performing the self-attention operation on ASes, the correlation coefficient $e_{ij}^{(k)}$ between the two ASes is obtained, as shown in Equation (1),

$$e_{ij}^{(k)} = \text{LeakyReLU}\left(\mathbf{a}^{(k)T} \left[\mathbf{W}^{(k)} z_i^{(k-1)} \parallel \mathbf{W}^{(k)} z_j^{(k-1)} \right]\right) \quad (1)$$

where $\mathbf{a}^{(k)T}$ is a learnable weight vector of the k^{th} encoder layer, and $z_i^{(k-1)}$ is the node embedding of AS i output from the $(k-1)^{\text{th}}$ encoder layer. $z_i^{(0)} = x_i$ denotes the initial node feature.

It is necessary to use the same scale when comparing the correlation coefficients between ASes. Therefore, the correlation coefficient is normalized using the Softmax function, as shown in Equation (2). The complete method for calculating attention coefficients is shown in Equation (3).

$$\alpha_{ij}^{(k)} = \text{Softmax}\left(e_{ij}^{(k)}\right) = \frac{\exp\left(e_{ij}^{(k)}\right)}{\sum_{l \in N_i} \exp\left(e_{il}^{(k)}\right)} \quad (2)$$

$$\alpha_{ij}^{(k)} = \frac{\exp\left(\text{LeakyReLU}\left(\mathbf{a}^{(k)T} \left[\mathbf{W}^{(k)} z_i^{(k-1)} \parallel \mathbf{W}^{(k)} z_j^{(k-1)} \right]\right)\right)}{\sum_{l \in N_i} \exp\left(\text{LeakyReLU}\left(\mathbf{a}^{(k)T} \left[\mathbf{W}^{(k)} z_i^{(k-1)} \parallel \mathbf{W}^{(k)} z_l^{(k-1)} \right]\right)\right)} \quad (3)$$

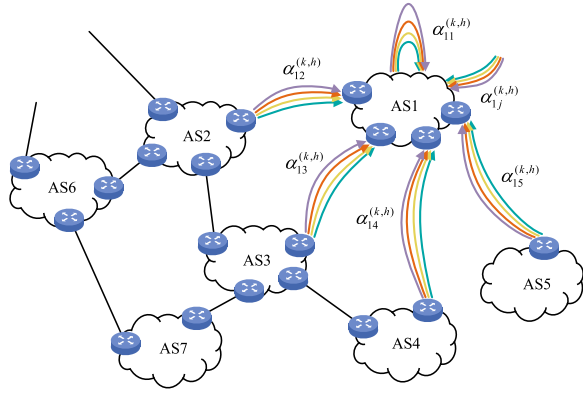


Fig. 5. Illustration of the multi-head attention mechanism ($H = 4$). AS1 updates its node embedding within neighbor ASes using the multi-head attention mechanism, where different colors represent different attention heads.

The new node embedding of AS i is computed using the learned attention coefficients. The AS embedding output by the k^{th} graph attention layer in the encoder is shown in Equation (4). The graph representation of AS-level topology is denoted as $Z^{(k)} = \{z_1^{(k)}, z_2^{(k)}, \dots, z_N^{(k)}\}$,

$$z_i^{(k)} = \sum_{j \in N_i} \alpha_{ij}^{(k)} \sigma(\mathbf{W}^{(k)} z_j^{(k-1)}) \quad (4)$$

where σ denotes a nonlinear activation function.

The multi-head attention mechanism refers to the use of H groups of mutually independent attention heads, where the node embeddings output by each attention head are either concatenated or averaged to obtain final node embedding, as shown in Fig. 5. The multi-head attention mechanism enables the graph attention layer to capture richer network information and reduce random noise interference.

The graph attention layer constructed in this paper computes multiple independent attention heads simultaneously, thereby enhancing the representational capacity of the model. The output AS embeddings are averaged, as shown in Equation (5).

$$z_i^{(k)} = \frac{1}{H} \sum_{h=1}^H \sum_{j \in N_i} \alpha_{ij}^{(k,h)} \sigma(\mathbf{W}^{(k,h)} z_j^{(k-1)}) \quad (5)$$

2) *Decoder Architecture:* Since the graph representation of AS-level topology already contains node attributes of ASes and network structural information, an inner product decoder is employed to reconstruct the adjacency matrix. This approach is simple and efficient, with the computation process given in Equation (6),

$$\hat{A} = \sigma(Z Z^T) \quad (6)$$

where, σ represents the Sigmoid activation function.

3) *Loss Function:* The training objective of the graph attention auto-encoder model is to minimize the reconstruction error of the adjacency matrix. In this paper, binary cross-entropy [18] is used as the loss function to measure the

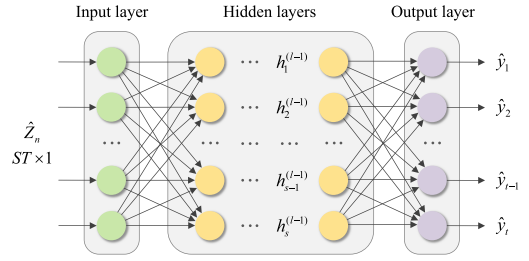


Fig. 6. The architecture of the multi-layer perceptron model. The model consists of input, hidden, and output layers, each layer fully connected to the next. The activation function of hidden layers is ReLU, while the activation function of the output layer is Softmax.

difference between the real adjacency matrix and reconstructed adjacency matrix, as expressed in Equation (7),

$$L_{GATE} = -\frac{1}{E_p} \sum_{i=1}^N \sum_{j=1}^N A_{ij} \log(\hat{A}_{ij}) - \frac{1}{E_n} \sum_{i=1}^N \sum_{j=1}^N (1 - A_{ij}) \log(1 - \hat{A}_{ij}) \quad (7)$$

where, E_p represents the number of positive edges in AS-level topology, $E_p = \sum_{i=1}^N \sum_{j=1}^N A_{ij}$, and E_n represents the number of negative edges in AS-level topology, $E_n = \sum_{i=1}^N \sum_{j=1}^N (1 - A_{ij})$.

D. Large-Scale BGP Routing Anomaly Detection Based on Multi-Layer Perceptron

The pre-trained graph attention auto-encoder model has learned graph representations of AS-level topologies without large-scale anomalies. The AS-level topology graph sequence is input into the encoder of the pre-trained model to generate AS embeddings. Since there is a significant difference between the embeddings of normal ASes and those with anomalies, a classifier is utilized to detect large-scale BGP routing anomalies. The multi-layer perceptron [16], [18] can better classify AS embeddings that contain nonlinear relationships through a multi-layer neural network structure. Therefore, the multi-layer perceptron is selected as the classifier. The architecture of the constructed multi-layer perceptron and the loss function are described as follows.

1) *Multi-Layer Perceptron Architecture:* The architecture of the multi-layer perceptron model constructed in this paper is shown in Fig. 6. The AS embedding is flattened into a one-dimensional vector and passed through the input layer to hidden layers. Each neuron in hidden layers performs weight and bias operations, as shown in Equation (8), followed by a nonlinear transformation using an activation function. The output is passed to the next hidden layer or the output layer, which generates the prediction probabilities of the normal and abnormal classes. The class with the highest probability is selected as the final detection result, as shown in Equation (9),

$$h_i^{(l)} = \sum_{j=1}^s w_{ij}^{(l-1)} h_j^{(l-1)} + b_i^{(l-1)} \quad (8)$$

$$\hat{y} = \arg \max_k \hat{y}_k \quad (9)$$

where $h_i^{(l)}$ represents the hidden layer output, $w_{ij}^{(l-1)}$ is the weight connecting different neurons, and $b_i^{(l-1)}$ is the bias term.

2) *Loss Function*: The binary cross-entropy is adopted as the loss function, as shown in Equation (10), to compute the error between the anomaly detection result and the true class. The error is backpropagated from the output layer to hidden layers to update model parameters, including weights and biases,

$$L_{MLP} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (10)$$

where N denotes the number of samples, y_i represents the true class of the sample, and $\hat{y}_i$ refers to the large-scale BGP routing anomaly detection result output by the multi-layer perceptron model.

E. Effectiveness Analysis

We evaluate the proposed method's effectiveness from the following two aspects: the effectiveness of obtaining AS embeddings based on a graph attention auto-encoder and the effectiveness of learning normal BGP routing behavior in an unsupervised manner.

1) *Obtaining AS Embeddings Based on a Graph Attention Auto-Encoder*: Existing large-scale BGP routing anomaly detection methods based on graph representation learning aggregate node embeddings of neighbor ASes using averaging or summing techniques. However, Comarela and Crovella [19] analyzed a large number of historical BGP data and found that ASes triggering large-scale BGP routing anomalies are typically those with larger network scales and significant positions in the network. We collect all routing information tables [44], [45], [46] from December 1, 2024, to extract AS paths. We preprocess the path information and then construct the Internet AS-level topology graph to calculate the degrees of ASes. The degree and ASRank¹ (ranked by the number of direct and indirect customers) of the ASes that trigger large-scale BGP routing anomalies are shown in Tab. II. AS174, AS12389, and other ASes exhibit large degrees and rank highly in ASRank. In contrast, the average degree of the topology is 5.53. This suggests that existing methods, which aggregate AS neighbor information indiscriminately to obtain AS embeddings, involve many non-important or even irrelevant neighbor ASes, affecting the effectiveness of large-scale BGP routing anomaly detection.

We introduce an attention mechanism into graph auto-encoder, learning AS embeddings using a graph attention auto-encoder. The proposed graph attention auto-encoder consists of an encoder and a decoder. The encoder computes attention coefficients α between each AS and its neighbor nodes based on the scale and network structure of ASes, assigning higher attention to important ASes that are more likely to cause large-scale BGP routing anomalies. The encoder outputs a low-dimensional graph representation of the AS-level topology. The decoder reconstructs the adjacency

TABLE II
THE DEGREE AND ASRANK OF ASes THAT TRIGGER LARGE-SCALE BGP ROUTING ANOMALIES

Ranked by anomaly events number	Number	Name	Degree	ASRank
1	AS174	COGENT-174	6782	3
2	AS9498	BBIL-AP	1206	23
3	AS9002	RETN-AS	2174	12
4	AS3356	LEVEL3	6658	1
5	AS6939	HURRICANE	9323	5
6	AS3549	LVL-3549	612	71
7	AS12389	ROSTELECOM-AS	1248	18

matrix $\hat{A}$ from the graph representation of AS-level topology. The graph attention auto-encoder model learns a graph representation that contains rich information by minimizing the error between the true and reconstructed adjacency matrices, thereby more accurately capturing the state of AS-level topology.

Overall, it can be concluded that obtaining AS embeddings based on a graph attention auto-encoder is effective.

2) *Learning Normal BGP Routing Behavior in an Unsupervised Manner*: Machine learning-based methods consider large-scale BGP routing anomaly detection as a binary classification problem. The detection result $\hat{y} \in \{0, 1\}$. Potential anomalies are detected by learning the differences between the AS embeddings under normal conditions and those during large-scale BGP routing anomalies. Existing methods typically rely on supervised learning to learn BGP routing behaviors, requiring labeled anomaly data. However, BGP anomaly data is insufficient, and the accuracy of labeled anomaly data cannot be guaranteed. Obtaining a large and accurate labeled dataset of BGP anomaly events is challenging, which impacts the effectiveness of large-scale BGP routing anomaly detection.

We mine a large amount of unlabeled normal BGP data using a graph attention auto-encoder. The model is pre-trained in an unsupervised manner to learn normal BGP routing behaviors. When large-scale BGP routing anomalies occur in the Internet, ASes exhibit anomalous routing behaviors, with significant changes in their connection relationships with other ASes. The pre-trained graph attention auto-encoder model cannot accurately reconstruct such anomalous AS-level topologies, and the output AS embeddings show significant differences from the embeddings of normal AS-level topologies. As shown in Fig. 7, for different large-scale BGP routing anomaly events, the reconstruction error for normal AS-level topologies is small and remains stable, while the reconstruction error for abnormal AS-level topologies is large and fluctuates significantly.

Overall, it can be concluded that learning normal BGP routing behavior in an unsupervised manner is effective.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

To verify the effectiveness of the proposed method, we conduct detection experiments on a dataset consisting of four

¹<https://asrank.caida.org/>

TABLE III
EXPERIMENTAL SETTINGS

Dataset		Parameter settings			Comparative methods	Evaluation metrics
Anomaly events	Type	Model	Parameter	Value		
TTNet	Route leaks	GATE	Learning rate	0.001	<i>BSVM</i> [10], <i>BLSTM</i> [11],	Accuracy
			Batch size	16		Precision
IndoSat	Prefix hijacking	GATE	Attention heads	4	<i>BMLP</i> [12], <i>BGNN</i> [16],	Recall
			Embedding dimension	16		F1-score
Rostelecom	Prefix hijacking	MLP	Epoch	100	<i>BGIN</i> [17], <i>BGAE</i> [18]	FPR
			Learning rate	0.001		
India	Route leaks	MLP	Output dimension of input layer, hidden layers, and output layer	64, 32, 1		
			Epoch	100		

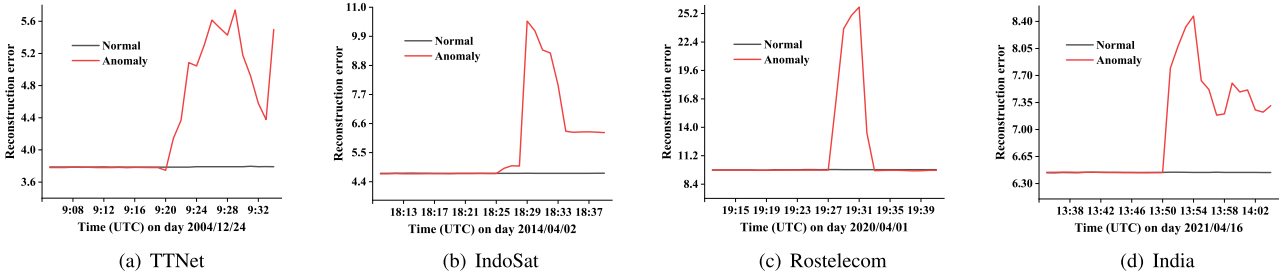


Fig. 7. Comparison of reconstruction errors between AS-level topologies with and without anomalies. During anomalies, ASes exhibit abnormal routing behaviors. The pre-trained model fails to accurately learn such AS embeddings, and the reconstruction error of the AS-level topology increases rapidly and becomes unstable.

large-scale BGP routing anomaly events. The experiments were executed on a workstation equipped with an Intel Xeon Gold 6248 CPU, 256 GB RAM, and NVIDIA RTX 4090. All methods were implemented with Pytorch geometric library and scikit-learn library. The proposed method is compared with existing typical detection methods, and its superiority is analyzed.

A. Experimental Settings

The experimental settings in this paper are shown in Tab. III, including the dataset, parameter settings, comparative methods, and evaluation metrics.

1) *Dataset*: We construct the dataset using large-scale BGP routing anomaly events (TTNet, IndoSat) adopted by existing typical detection studies [11], [12], [16], as well as recent large-scale BGP routing anomaly events (Rostelecom, India). The detailed information of anomaly events is shown in Tab. IV.

Existing studies widely utilize publicly available BGP data to analyze anomaly events. The BGP data collection projects, including RouteViews² and RIPE RIS,³ deploy route collectors around the world to obtain real-time BGP data. We utilize the BML tool [47] to automatically collect routing information tables and UPDATE messages about historical anomaly events,

TABLE IV
LARGE-SCALE BGP ROUTING ANOMALY EVENTS

Events	Type	Start time	Duration (min)	Origin
TTNet	Route leaks	2004/12/24 09:20 (UTC)	630	AS9121
IndoSat	Route hijacking	2014/04/02 18:25 (UTC)	150	AS4761
Rostelecom	Route hijacking	2020/04/01 19:27 (UTC)	5	AS12389
India	Route leaks	2021/04/16 13:50 (UTC)	10	AS55410

containing 3,038,314 routing announcement records. The AS paths are extracted to construct AS-level topology graphs, which are used as positive samples. Routing information tables and UPDATE messages are collected one day before historical anomaly events, containing 1,539,070 routing announcement records. We manually check this routing data based on three authoritative sources, including Oracle blogs,⁴ BGPStream,⁵ and BGPmon.⁶ During these periods, no large-scale BGP routing anomalies are reported. Therefore, we extract AS path information to construct AS-level topology graphs, which are used as negative samples. There are 8 samples, with a ratio of positive to negative samples of 1:1. For different samples, the total duration of BGP data collection is 1 hour. To balance the detection accuracy and inference time, we

²<https://www.routeviews.org/routeviews/>

³<https://www.ripe.net/ris/>

⁴<https://blogs.oracle.com/>

⁵<https://bgpstream.ca/ida.org/>

⁶ <https://www.bgpmon.net/>

TABLE V
RESULTS OF LARGE-SCALE BGP ROUTING ANOMALY DETECTION

Anomaly events	Evaluation metrics	Methods based on manual feature extraction			Methods based on graph representation learning			Ours
		<i>BSVM</i> [10]	<i>BLSTM</i> [11]	<i>BMLP</i> [12]	<i>BGNN</i> [16]	<i>BGIN</i> [17]	<i>BGAE</i> [18]	
TTNet	Accuracy	89.0%	96.0%	93.0%	97.0%	96.0%	100.0%	<u>100.0%</u>
	Precision	89.8%	100.0%	97.8%	96.1%	96.0%	100.0%	<u>100.0%</u>
	Recall	88.0%	92.0%	88.0%	98.0%	96.0%	100.0%	<u>100.0%</u>
	F1 score	88.9%	95.8%	92.6%	97.0%	96.0%	100.0%	<u>100.0%</u>
	FPR	10.0%	0.0%	2.0%	4.0%	4.0%	0.0%	<u>0.0%</u>
IndoSat	Accuracy	87.0%	94.0%	91.0%	97.0%	93.0%	100.0%	<u>100.0%</u>
	Precision	87.8%	97.8%	91.8%	96.1%	92.2%	100.0%	<u>100.0%</u>
	Recall	86.0%	90.0%	90.0%	98.0%	94.0%	100.0%	<u>100.0%</u>
	F1 score	86.9%	93.7%	90.9%	97.0%	93.1%	100.0%	<u>100.0%</u>
	FPR	12.0%	2.0%	8.0%	4.0%	8.0%	0.0%	<u>0.0%</u>
Rostelecom	Accuracy	80.0%	92.0%	86.0%	91.0%	89.0%	90.0%	<u>93.0%</u>
	Precision	81.2%	95.7%	87.5%	90.2%	89.8%	91.7%	<u>95.7%</u>
	Recall	78.0%	88.0%	84.0%	<u>92.0%</u>	88.0%	88.0%	90.0%
	F1 score	79.6%	91.7%	85.7%	91.1%	88.9%	89.8%	<u>92.8%</u>
	FPR	18.0%	4.0%	12.0%	10.0%	10.0%	8.0%	<u>4.0%</u>
India	Accuracy	84.0%	93.0%	90.0%	94.0%	89.0%	95.0%	<u>96.0%</u>
	Precision	85.4%	<u>97.8%</u>	90.0%	92.3%	88.2%	95.9%	96.0%
	Recall	82.0%	88.0%	90.0%	96.0%	90.0%	94.0%	<u>96.0%</u>
	F1 score	83.7%	92.7%	90.0%	94.1%	89.1%	94.9%	<u>96.0%</u>
	FPR	14.0%	<u>2.0%</u>	10.0%	8.0%	12.0%	4.0%	4.0%
Average	Accuracy	85.0%	93.8%	90.0%	94.8%	91.8%	96.3%	<u>97.3%</u>
	Precision	86.1%	97.8%	91.8%	93.7%	91.6%	97.0%	<u>98.0%</u>
	Recall	83.5%	89.5%	88.0%	96.0%	92.0%	95.5%	<u>96.5%</u>
	F1 score	84.8%	93.5%	89.8%	94.8%	91.8%	96.2%	<u>97.2%</u>
	FPR	13.5%	2.0%	8.0%	6.5%	8.5%	3.0%	<u>2.0%</u>

extract AS-level topology graphs with a two-minute interval. Each sample consists of 30 AS-level topology graphs, $G = \{G_1, G_2, \dots, G_{30}\}$.

2) *Parameter Settings*: The model performance is evaluated using leave-one-out cross-validation. One anomaly event from the dataset is used as the test set, while the remaining anomaly events serve as the training set. This process is repeated, with each anomaly event used as the test set. The final performance evaluation is the average of experimental results across all test sets. To minimize the impact of randomness, we repeated the experiment 50 times. Grid search is employed to tune model hyperparameters. The optimal parameters for the Graph Attention Auto-Encoder (GATE) model are a learning rate of 0.001, batch size of 16, attention heads of 4, node embedding dimension of 16, and training epochs of 100. The optimal parameters for the Multi-layer Perceptron (MLP) model are a learning rate of 0.001, output feature dimensions of 64, 32, and 1 for the input, hidden, and output layers, respectively, and training epochs of 100.

B. Comparison of Detection Performance

To verify the effectiveness of the proposed method, we compare it with typical detection methods based on manual feature extraction, including *BSVM* [10], *BLSTM* [11], and *BMLP* [12], as well as typical detection methods based on graph representation learning for feature extraction, including *BGNN* [16], *BGIN* [17], and *BGAE* [18].

The proposed method and existing typical detection methods are used to detect large-scale BGP routing anomaly events. The experimental results are shown in Tab. V. The best result for each evaluation metric is highlighted in bold and underlined. The confusion matrix is shown in Fig. 8. For different large-scale BGP routing anomaly events, the proposed method achieves average values of accuracy, precision, recall, F1 score, and FPR of 97.3%, 98.0%, 96.5%, 97.2%, and 2.0%, respectively, outperforming all existing typical detection methods. The proposed method achieves the lowest FPR, indicating that it can effectively reduce false alarms while maintaining high detection accuracy. The experimental results show that the proposed method is effective. Utilizing a graph attention auto-encoder can generate AS embeddings with richer information, thereby enhancing the detection performance of large-scale BGP routing anomalies.

However, existing typical methods perform poorly in detecting recent anomaly events, including Rostelecom and India. The proposed method also has misclassifications. This can be attributed to the continuous expansion of Internet AS-level topology, with a significant increase in the number of nodes and edges. During the TTNet anomaly event on December 24, 2004, there were 18,780 ASes and 69,691 edges. During the India anomaly event on April 16, 2021, the number of ASes increased to 71,387 and the number of edges increased to 322,593. As the network becomes increasingly complex, interference from non-important, even irrelevant

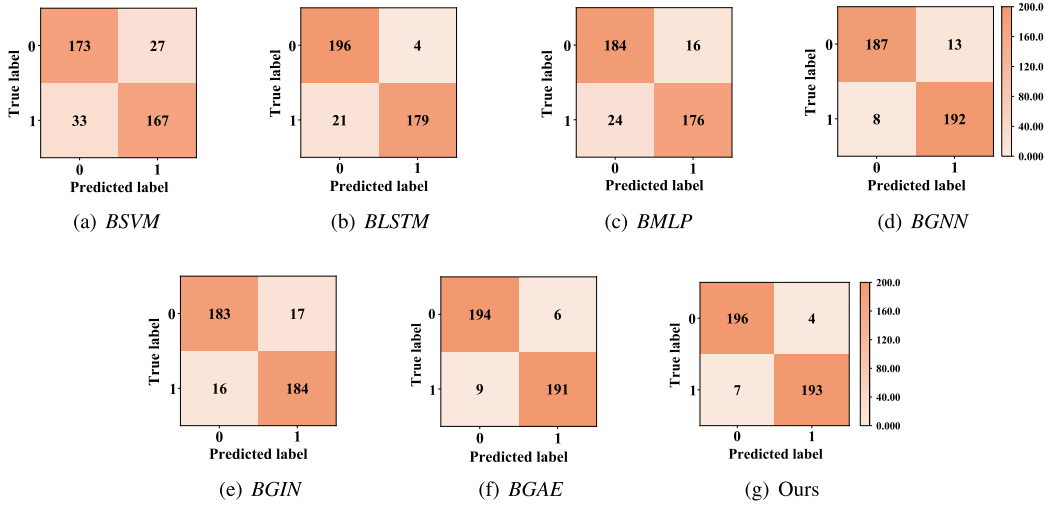


Fig. 8. Confusion matrix heatmap. The figure demonstrates the detection performance of different methods on multiple anomaly event datasets.

ASes makes large-scale BGP routing anomaly detection more challenging.

C. Attention Coefficient Analysis

We present a statistical analysis of attention coefficients learned by the constructed graph attention auto-encoder model and compare them with the same weights assigned to neighbor ASes.

The entropy of the attention coefficients is computed to quantify their distribution, as shown in Equation (11). A lower entropy value indicates a highly concentrated distribution of attention coefficients, with an entropy of 0 signifying that the attention is entirely focused on a single AS. A higher entropy value indicates a more dispersed distribution of attention coefficients, with the highest entropy $\log(N_i)$ occurring in the case of a uniform distribution,

$$H(\{\alpha_{ij}\}_{j \in N(i)}) = - \sum_{j \in N(i)} \alpha_{ij} \log(\alpha_{ij}) \quad (11)$$

where N_i is the number of neighbor nodes of AS i .

For different large-scale BGP routing anomaly events, the entropy distribution of attention coefficients is statistically analyzed to produce a Cumulative Distribution Function (CDF) curve, as shown in Fig. 9. The attention coefficient entropy values learned by the proposed graph attention auto-encoder model are predominantly concentrated between 0 and 0.5, significantly higher than those under a uniform distribution. The model exhibits a lower entropy distribution of attention coefficients, suggesting that it focuses more on certain important ASes, thereby improving the accuracy of large-scale BGP routing anomaly detection.

D. Representational Capacity Evaluation

We evaluate the representational capacity of the proposed method from two perspectives: proximity and hierarchy.

1) *AS Proximity*: We use the similarity metric $s(v_i, v_j)$ based on Euclidean distance $d_E(z_{v_i}, z_{v_j})$ to evaluate the embedding differences between adjacent and non-adjacent

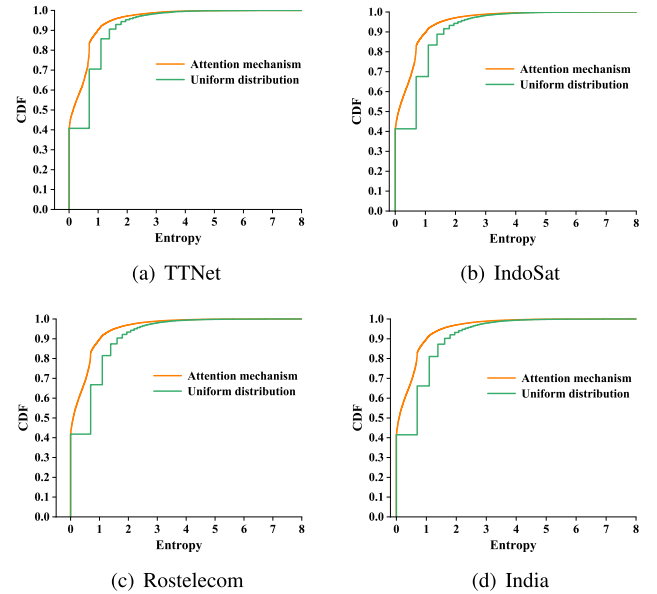


Fig. 9. Comparison of AS attention coefficient entropy distributions. We compare the entropy of attention coefficients learned by the proposed method with a uniform distribution, where the same weights are assigned to all neighboring ASes.

ASes, as shown in Equations (12) and (13). The set of adjacent ASes with different proportions of common neighbors is denoted as $E_{\text{adj}} = \{(v_i, v_j) \mid \frac{|N(v_i) \cap N(v_j)|}{|N(v_i) \cup N(v_j)|} \geq \tau\}$, and the set of non-adjacent ASes is denoted as $E_{\text{non-adj}} = \{(v_i, v_j) \mid v_i \neq v_j \wedge (v_i, v_j) \notin E\}$.

$$d_E(z_{v_i}, z_{v_j}) = \|z_{v_i} - z_{v_j}\|_2 = \sqrt{\sum_{k=1}^d (z_{v_i,k} - z_{v_j,k})^2} \quad (12)$$

$$s(u, v) = \frac{1}{1 + d_E(z_{v_i}, z_{v_j})} \quad (13)$$

The adjacent AS sets are constructed with common neighbor proportions more than 25%, 50%, and 75%, respectively. Five thousand non-adjacent ASes are randomly sampled to form the

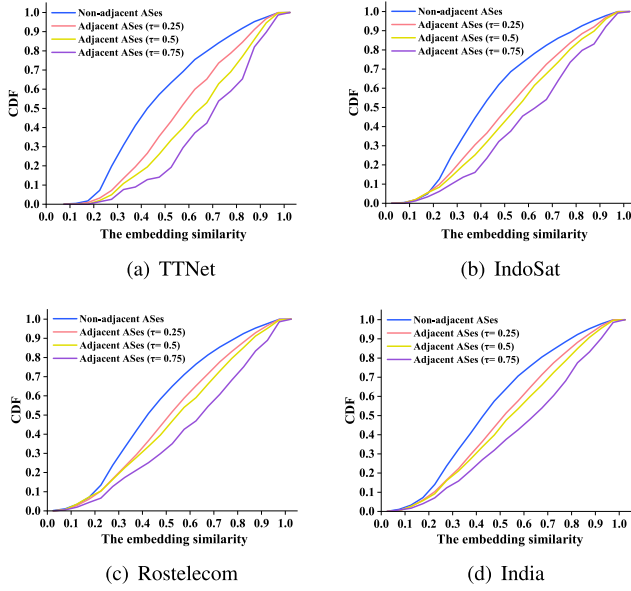


Fig. 10. CDF of node embedding similarity for adjacent and non-adjacent ASes. For adjacent ASes, the proportion of common neighbors is set to 0.25, 0.5, and 0.75, respectively.

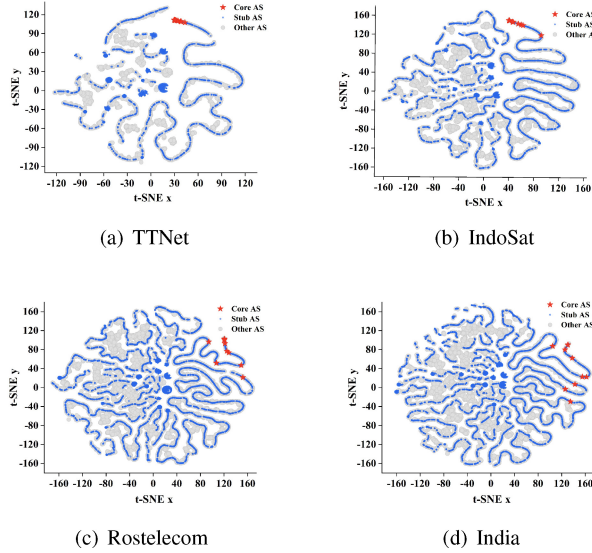


Fig. 11. The results of dimensionality reduction visualization for different anomaly events via t-SNE. There is a significant difference in the distribution of node embeddings between core ASes and stub ASes.

non-adjacent set. The embedding similarity between AS pairs within the adjacent and non-adjacent AS sets is calculated, and the CDF of similarity is shown in Fig. 10. The experimental results indicate that the embedding similarity of adjacent ASes is significantly higher than that of non-adjacent ASes. Furthermore, as the proportion of common neighbors increases (i.e., the topological structure becomes more similar), the embedding similarity of adjacent ASes increases. These findings demonstrate that the proposed method effectively captures the proximity between ASes.

2) *AS Hierarchy*: The Internet AS-level topology exhibits a hierarchical structure. Core ASes are located at the high tier of the Internet and form the backbone of global connectivity,

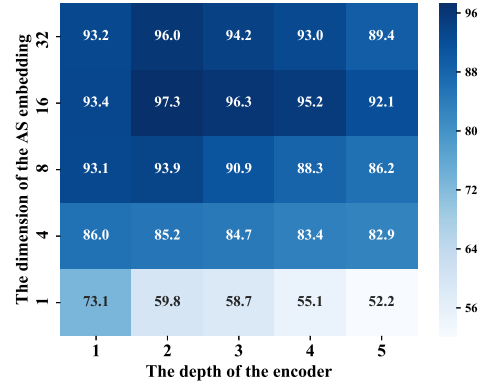


Fig. 12. Detection accuracy using different parameter combinations. The detection accuracy is highest when $L = 2$, $S = 16$.

including Tier 1 Internet service providers, whose degrees are significantly higher than other ASes. Conversely, stub ASes are located at the low tier of the Internet, accessing the Internet via core ASes or their downstream networks. To verify the representational capacity of AS hierarchy, we utilize the t-SNE method [48] to reduce the high-dimensional AS embedding to two dimensions for visual analysis.

The results of t-SNE dimensionality reduction visualization for different anomaly events are shown in Fig. 11. Core ASes (the top 10 ASes by degree), stub ASes (ASes with a degree of 1), and other ASes are marked with different colors. It can be observed that the node embeddings of core ASes are relatively concentrated in distribution, exhibiting high similarity. In contrast, the node embeddings of stub ASes are widely dispersed, demonstrating significant heterogeneity. These results indicate that the proposed method effectively captures the hierarchical structure inherent in the Internet AS-level topology.

E. Parameter Sensitivity Analysis

We analyze the impact of key parameters of the graph attention auto-encoder on large-scale BGP routing anomaly detection performance.

The key parameters of the graph attention auto-encoder include the depth of the encoder L and the dimension of the AS embedding S . The encoder takes the AS embedding as input and outputs the transformed embedding. The depth of the encoder determines the scope of information propagation. The AS embedding is a low-dimensional vector containing graph structural information, and its dimension influences the model's representational capacity.

We set $L = \{1, 2, 3, 4, 5\}$ and $S = \{1, 4, 8, 16, 32\}$. Large-scale BGP routing anomaly detection experiments are conducted using different parameter combinations. The detection accuracy is shown in Fig. 12. $L = 2$, $S = 16$ is the optimal parameter combination, with a detection accuracy of 97.3%. The experimental results indicate that properly increasing the depth of the encoder can improve detection accuracy. But over-smoothing occurs when the depth of the encoder is too large, causing the node embeddings of different ASes to tend to be the same. Conversely, if it is too large, it may include

TABLE VI
ABLATION EXPERIMENT RESULTS COMPARISON

Embedding acquisition		Embedding classification			Training strategy		Detection performance				
GAE	GATE	SVM	FC	MLP	Joint training	Phased training	Accuracy	Precision	Recall	F1 score	FPR
✓	-	✓	-	-	✓	-	90.3%	89.7%	91.0%	90.3%	10.5%
✓	-	✓	-	-	-	✓	92.8%	92.5%	93.0%	92.8%	7.5%
✓	-	-	✓	-	✓	-	88.3%	88.4%	88.0%	88.2%	11.5%
✓	-	-	✓	-	-	✓	91.0%	91.0%	91.0%	91.0%	9.0%
✓	-	-	-	✓	✓	-	94.3%	94.9%	93.5%	94.2%	5.0%
✓	-	-	-	✓	-	✓	96.3%	97.0%	95.5%	96.2%	3.0%
-	✓	✓	-	-	✓	-	92.3%	92.0%	92.5%	92.3%	8.0%
-	✓	✓	-	-	-	✓	94.0%	93.6%	94.5%	94.0%	6.5%
-	✓	-	✓	-	✓	-	90.0%	90.4%	89.5%	90.0%	9.5%
-	✓	-	✓	-	-	✓	92.3%	92.5%	92.0%	92.2%	7.5%
-	✓	-	-	✓	✓	-	95.3%	95.5%	95.0%	95.3%	4.5%
-	✓	-	-	✓	-	✓	97.3%	98.0%	96.5%	97.2%	2.0%

noise and significantly increase computational resource consumption.

F. Ablation Studies

To evaluate the contribution of different components of the proposed method for large-scale BGP routing anomaly detection, we conduct ablation experiments on the attention mechanism, MLP model, and phased training strategy employed in the proposed method.

Graph auto-encoder and graph attention auto-encoder are selected to obtain AS embeddings, and these embeddings are classified using Support Vector Machine (SVM), Fully Connected Layer (FC), and MLP models, respectively. The models are trained with joint training and phase-based training strategies, respectively. The average prediction performance on different anomaly events for the combinations of models and training strategies is shown in Tab. VI.

In terms of embedding acquisition, the detection performance of the graph attention auto-encoder outperforms that of the graph auto-encoder. This indicates that by introducing the attention mechanism, the model can focus more on important ASes where large-scale BGP routing anomalies are likely to occur, thus improving the accuracy of anomaly detection. In terms of embedding classification, the MLP model performs significantly better than the SVM and FC models, capturing the complex nonlinear relationships in AS embeddings. In terms of training strategies, the detection accuracy is higher under the phase-based training strategy. The unsupervised loss aims to learn the features of the normal topology, while the supervised loss aims to distinguish anomalous states. There is some conflict between these two optimization objectives, and joint training in the second stage results in a decline in prediction performance.

G. Anomaly Origin AS Localization

Based on the differences between the AS embeddings during anomaly events and those in a normal network state, we provide a list of possible anomaly origin ASes.

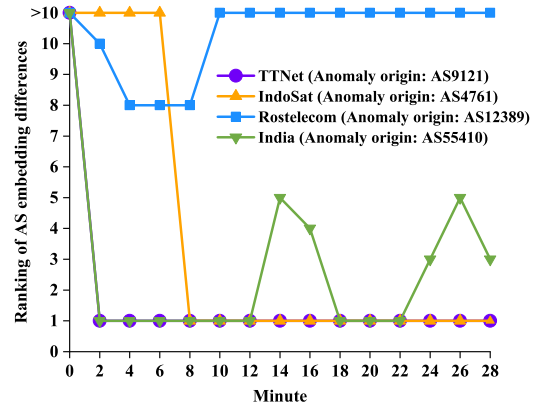


Fig. 13. Ranking of node embedding differences of anomaly origin ASes. The node embedding difference rankings of the anomaly origin ASes are all within the top 10 for different anomaly events.

TABLE VII
LIST OF POTENTIAL ANOMALY ORIGIN ASes

Anomaly event	Top 10 ASes with the largest node embedding differences
TTNet	<u>AS9121</u> , AS25219, AS25594, AS33910, AS6822, AS15897, AS20871, AS20954, AS8927, AS23242
IndoSat	<u>AS4761</u> , AS59134, AS45708, AS131743, AS56234, AS38523, AS23691, AS55694, AS58514, AS131757
Rostelecom	AS9185, AS47460, AS210160, AS208605, AS203994, AS201578, AS58178, <u>AS12389</u> , AS9121, AS202070
India	<u>AS55410</u> , AS133990, AS132343, AS135171, AS139553, AS133580, AS137640, AS137149, AS6852, AS133969

We use Euclidean distance to evaluate the differences in node embeddings of all ASes after an anomaly event occurs. The ranking of the differences in node embeddings for anomaly origin ASes is shown in Fig. 13. For the TTNNet anomaly event, the node embedding difference of the anomaly origin AS9121 consistently ranked first. For the IndoSat anomaly event, 8 minutes after the anomaly occurred, the node embedding difference of the anomaly origin AS4761 ranked first, indicating that the impact of the event spread gradually,

TABLE VIII
THE INFERENCE TIME OF DIFFERENT DETECTION METHODS

Anomaly events	Inference time of different methods (sec)						Ours
	<i>BSVM</i> [10]	<i>BLSTM</i> [11]	<i>BMLP</i> [12]	<i>BGNN</i> [16]	<i>BGIN</i> [17]	<i>BGAE</i> [18]	
TTNet	22.65	27.94	59.35	29.63	36.15	45.68	46.12
IndoSat	89.47	104.26	187.91	149.24	150.32	153.79	154.30
Rostelecom	156.42	175.63	298.67	232.29	235.84	240.17	240.68
India	173.96	193.28	326.59	272.69	276.53	281.35	281.79
Average	110.63	125.28	218.13	170.96	174.71	180.25	180.72

and the illegal routing went through a period of rerouting in the network. For the Rostelecom event, the node embedding difference of the anomaly origin AS12389 ranked within the top ten. Since the event lasted for a short time, the anomaly origin AS returned to normal after 10 minutes. For the India anomaly event, the node embedding difference of the anomaly origin AS55410 ranked within the top five.

Although not all anomaly events exhibit the largest node embedding differences for the anomaly origin ASes, in practical applications, the list of potential anomaly origin ASes can be inferred based on the node embedding differences, as shown in Tab. VII. The actual anomaly origin ASes are highlighted in bold and underlined, and their node embedding difference rankings are all within the top 10. This suggests that inferring potential anomaly origin ASes based on node embedding differences helps significantly narrow down the scope of anomaly origin AS investigations, supporting anomaly traceability and localization.

H. Analysis of Inference Times and Response Times

We compare the inference times of the proposed method and existing typical detection methods for different large-scale BGP routing anomaly events, and we analyze the response time of the proposed method.

1) *Inference Times*: Both the proposed method and existing typical methods consider large-scale BGP routing anomaly detection as a binary classification problem. They use machine learning to train detection models and deploy the trained models for anomaly detection, performing the inference process. Therefore, we compare the inference times for detecting anomalies on raw BGP data using the trained model in seconds, as shown in Tab. VIII. It can be seen that the proposed method does not provide the fastest inference time, but it is much faster than the *BMLP* method. This indicates that the proposed method sacrifices some time to improve detection accuracy. In the future, the lightweight large-scale BGP routing anomaly detection method will be the focus of our research.

2) *Response Times*: Additionally, we analyze the response time of the proposed method, which is the time interval from the occurrence of the anomaly to its successful detection by the model [16]. For different anomaly events, we sequentially collect BGP data within the time window $[t_0 - 30, t_0 + t]$, from 30 minutes before the anomaly event occurs to t minutes after its occurrence. $t = \{2, 4, 6, \dots, 30\}$. This data is used for

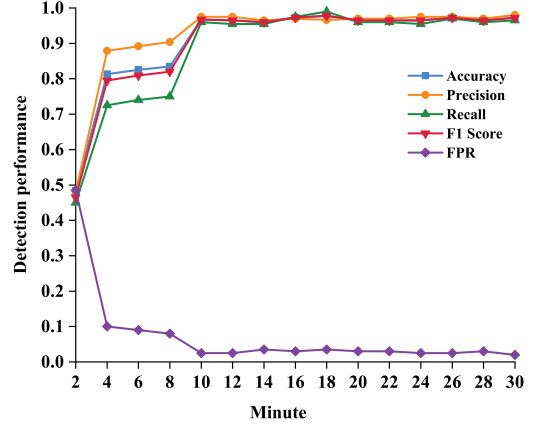


Fig. 14. Comparison of detection performance at different times after anomaly events occur. The detection performance gradually improves and then tends to stabilize.

model training and testing. The average detection performance under different time windows is shown in Fig. 14.

The experimental results indicate that 4 minutes after the anomaly event occurs, the accuracy, precision, recall, F1 score, and FPR are 81.3%, 87.9%, 72.5%, 79.5%, and 10.0%, respectively, achieving preliminary anomaly detection. Subsequently, the detection performance gradually improves. Ten minutes after the anomaly event occurs, the accuracy, precision, recall, F1 score, and FPR are 96.8%, 97.5%, 96%, 96.7%, and 2.5%, respectively, and the detection performance eventually stabilizes. This demonstrates that the proposed method is capable of detecting large-scale BGP anomalies in a short period. In the future, we will further investigate its real-time detection performance in practical deployment applications.

V. CONCLUSION

To solve the problems of interference from non-important ASes and the insufficiency of labeled anomaly data in existing typical detection methods, we propose a large-scale BGP routing anomaly detection method based on a graph attention auto-encoder. The proposed method pre-trains the graph attention auto-encoder model to obtain AS embeddings and classifies them based on the multi-layer perceptron to achieve large-scale BGP routing anomaly detection. The experimental results on four large-scale BGP routing anomaly event datasets show that the proposed method achieves higher detection accuracy than existing typical methods. In future work, the

lightweight anomaly detection method will be the focus of our research. And we will continue to evaluate our method by running it in a live deployment.

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